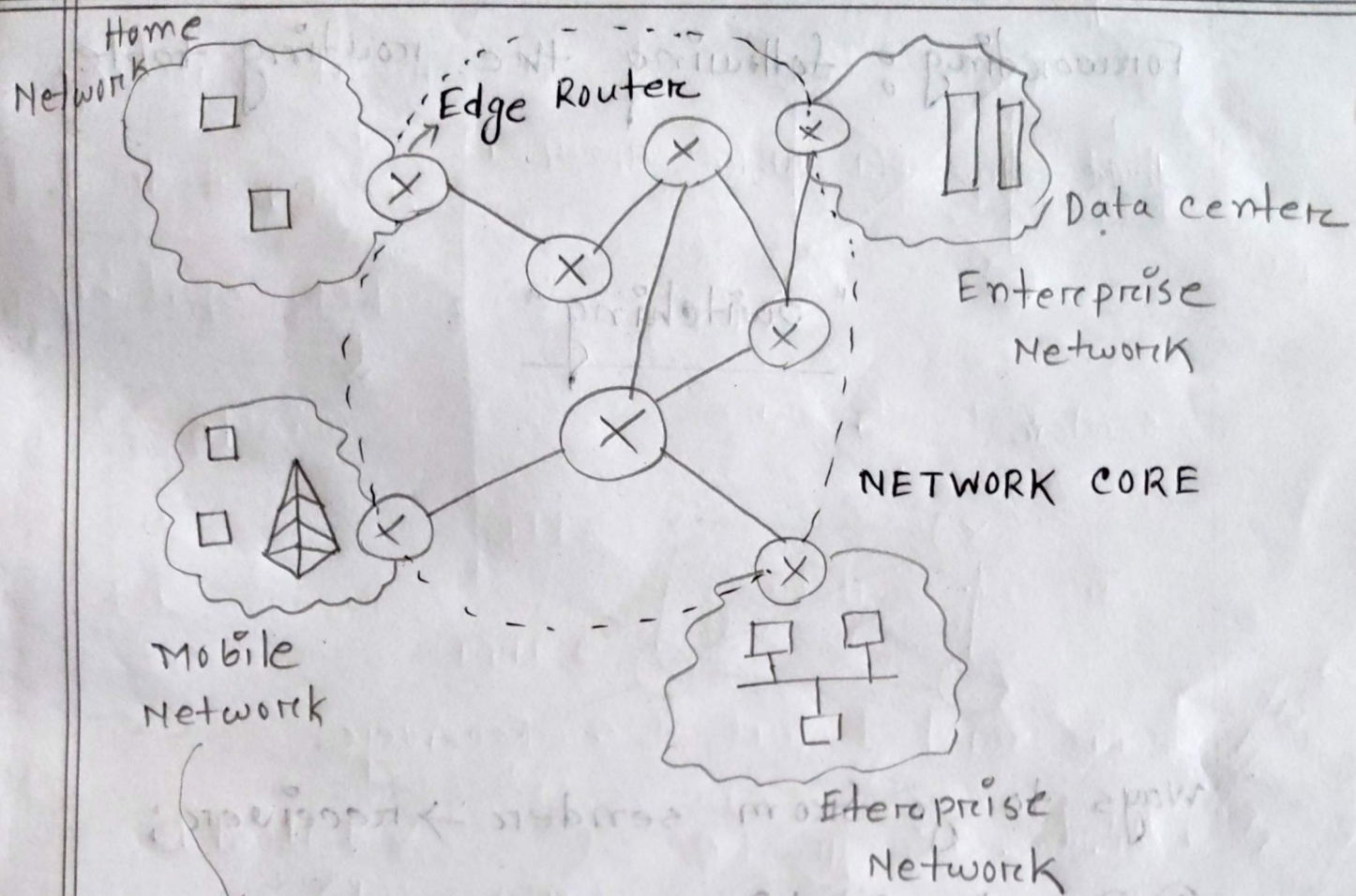


Computer Networks



- ① Edge Network
 - ② Network core
- ↓
- what connected
- } INTERNET

Protocols → set of rules

Routing & Forwarding / switching

- optimal path খুঁজে বের করে ।
 - routing table
 - Next কোনদিকে যাবে সেটা বলে দেয় ।
- } routing algorithm run করে পাঠায় ।

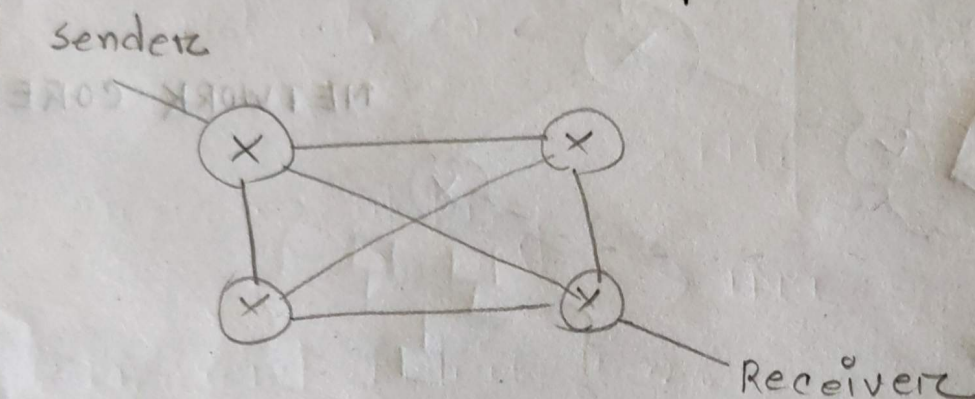
Q: Routing and forwarding difference.

action

Forwarding : following the routing table.

↓
table follow করে যাঠবে মাডুলা

"Switching"



ways to go from sender → receiver :

- ① Packet switching (Internet)
- ② circuit switching → fixed path থাকে

mid - - - >

diff

circuit

- ① Fixed path
- ② dedicated path

advantage : faster

disadvantage :

inefficient

old telephone

Packet

- ① variable path

advantage : efficient

disadvantage : significant delay

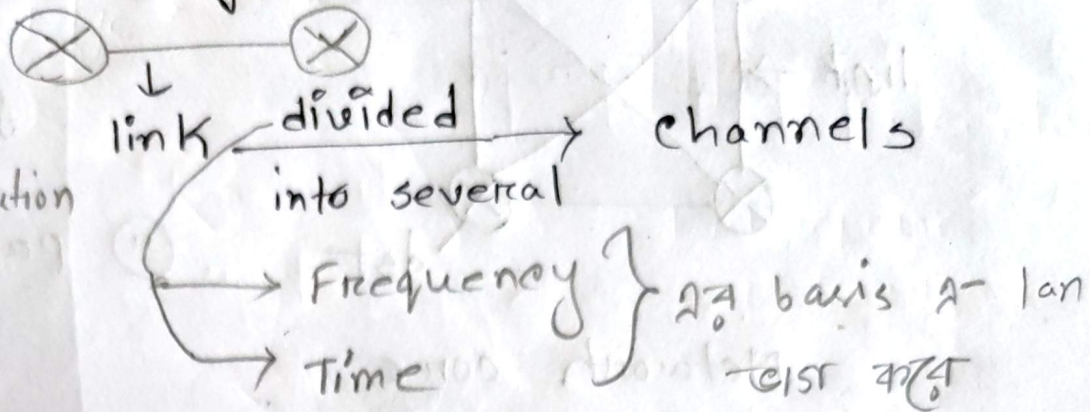
Internet

How circuit switching is done?

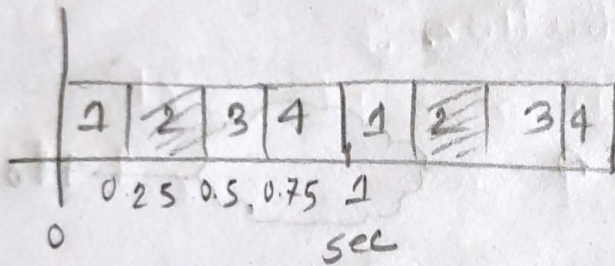
(FDM)
Frequency
Division $\downarrow$ Multiplication

180 - 200
160 - 180
140 - 160
120 - 140
100 - 120

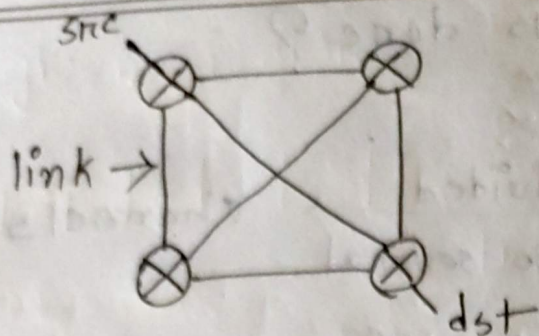
Ex: FM Radio



Time :



[TDM]



Network core

① circuit switching

↓
fixed + dedicated

② packet switching

↓
variable

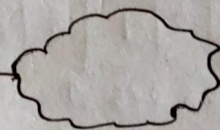
circuit switching

FDM

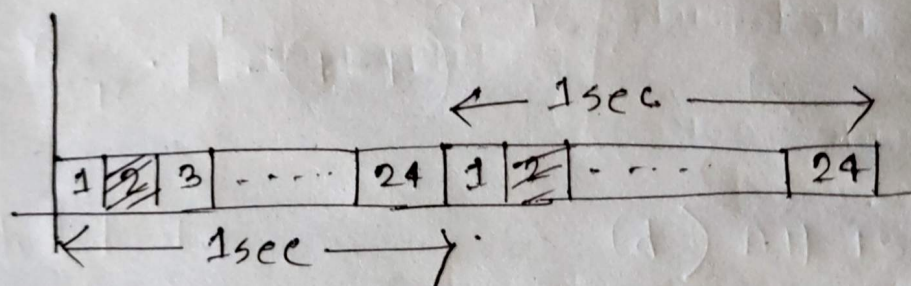
TDM

Exercise 4:

Host A



Host B



$$\hookrightarrow 1.536 \text{ Mbps} = 1.536 \times 10^6$$

$$\text{Allocated Bandwidth per slot} = \frac{1.536 \times 10^6}{24}$$

$$= 64000 \text{ bps}$$

$$\therefore \text{File transmission time} = \frac{640000}{64000}$$

$$= 10 \text{ s}$$

0 → bit
B → Byte
bit → 1 Byte

$$\therefore \text{Total time} = (10 + 0.5) = 10.5 \text{ sec}$$

↓
path create time.

Ex-5:

a) circuit: users = $\frac{2 \text{ Mbps}}{1 \text{ Mbps}}$

$$= 2$$

b) queuing delay for 2 users is

Ex-6:

a) no of channel / no of user = $\frac{200 \text{ Mbps}}{25 \text{ Mbps}}$

$$= 8$$

(b)

b) $15 > 8 \therefore \text{NO}$

worst

case :

15 are transmitting

wait  8 users

given/specific

☐ ☒ ☐

0.2 0.2 0.2 transmit

0.8 0.8 0.8 not u

$$P = 0.8 \times 0.2 \times 0.8$$

=

any : ৩৬(নতুন) ২৬-১৮৫

(i) ☒ ☐ ☐

$$= 0.2 \times 0.8 \times 0.8$$

(ii) ☐ ☒ ☐

$$= 0.8 \times 0.2 \times 0.8$$

(iii) ☐ ☐ ☒

$$= 0.8 \times 0.8 \times 0.2$$

Sum of all probabilities

Binomial Distribution:

trial $\rightarrow$ success/transmit $\rightarrow P$
 $\rightarrow$ fail/not u $\rightarrow 1-P$

$${}^n C_r \cdot P^r \cdot (1-P)^{n-r}$$

(c) total = 15

transmit = 1

Not u = 14

$$\text{specific} = 0.2 \times (0.8)^{14}$$

$$= 8.79 \times 10^{-3}$$

(d) All 15 users:

$$= (0.2)^{15}$$

=

(e) Generalised, $\sum_{i=0}^{15} {}^{15}C_i (0.2)^i (1-0.2)^{15-i}$

$\approx$ very small value

৩৬৬
more than
১৮৫/৩৬৬

Circuit switching

- Fixed + Dedicated Path
- Fixed num of users (no of channels)

Packet switching

- variable path
- can support more users than channels
- significant Delay.

Delay

depend
on
router

① Processing Delay → Fixed value

② Queuing

③ Transmission $u \rightarrow$ (Router $\rightarrow$ Medium)④ Propagation u (Medium)

$$\text{Total Delay, } D_{\text{total}} = d_{\text{pro}} + d_{\text{que}} + d_{\text{tran}} + d_{\text{prop}}$$

☐ Transmission Delay (d_{tran}) = $\frac{L}{R}$

$L \rightarrow$ packet size

$R \rightarrow$ link rate / link capacity / transmission speed
/ transmission rate

☐ Propagation Delay, $d_{\text{prop}} = \frac{d}{s}$

$d \rightarrow$ distance

$s \rightarrow$ medium speed

☐ Queuing Delay, $d_{\text{que}} = n * \frac{L}{R}$

Ex-8 :

Link 1 : $\cancel{d_{proc}} + \cancel{d_{que}} + d_{tran} + d_{prop}$

$$= 0 + 0 + \frac{16000}{1000 \times 10^6} + \frac{2 \times 10^3}{3 \times 10^8}$$

=

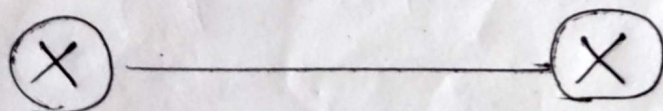
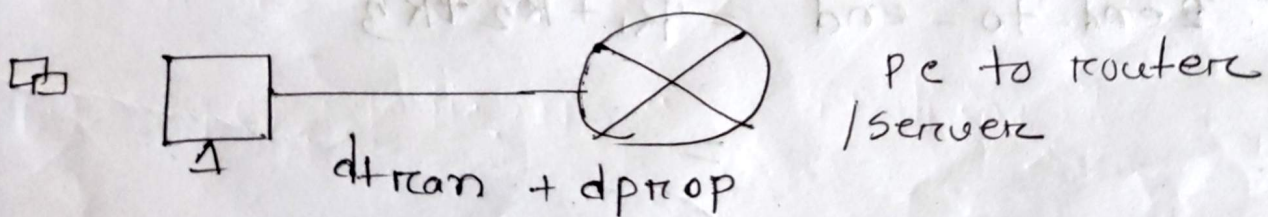
Link 2 : $0 + 0 + \frac{16000}{10 \times 10^6} + \frac{500 \times 10^3}{3 \times 10^8}$

=

Link 3 : $0 + 0 + \frac{16000}{100 \times 10^6} + \frac{3 \times 10^3}{3 \times 10^8}$

=

End-to-end = sum of all link.



$$d_{proc} + d_{que} + d_{tran} + d_{prop}$$

$$Gbps \rightarrow 10^9$$

$$\mu \rightarrow 10^{-6}$$

Ex-9:

$$R_1 = d_{tran} + d_{prop}$$

$$= \frac{L}{R} + \frac{d}{S}$$

$$= \frac{1 \times 10^3 \times 8}{1 \times 10^6} + \frac{100 \times 10^3}{2000 \times 10^3} =$$

$$R_2 = 5 \times 10^{-6} + \frac{1 \times 10^3 \times 8}{1 \times 10^6} + \frac{100 \times 10^3}{2000 \times 10^3}$$

=

$$R_3 = 5 \times 10^{-6} + \frac{1 \times 10^3 \times 8}{1 \times 10^6} + \frac{100 \times 10^3}{2000 \times 10^3}$$

=

$$\therefore \text{end-to-end} = R_1 + R_2 + R_3$$

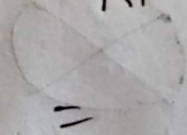
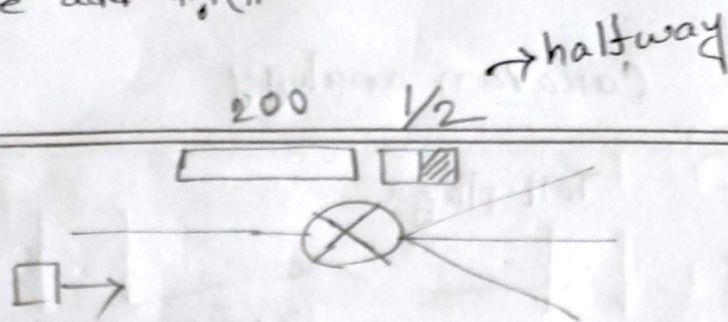


Figure add 4.4(4)

Q.4:



$$\text{queuing delay} = (200 + 0.5) * \frac{L}{R}$$

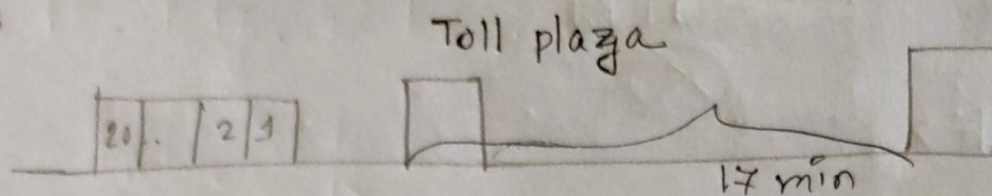
$$= 200.5 * \frac{64 \times 8}{10 \times 10^6}$$

=

practice - knowledge Test 2 → 3, 4, 5, 6, 7

Caravan analogy

Fall-253



cars = data packets

toll plaza = routers

toll time = transmission delay

travel time = propagation delay

$$(i) \quad d_{trans} = 2 \text{ min/per car}$$

$$= 20 * 2 = 40 \text{ min}$$

$$d_{prop} = \frac{d}{s} = \frac{500 \times 10^3}{2000 \times 10^3}$$

$$= 0.25 \text{ hr}$$

$$= \underline{15 \text{ min}}$$

$$(ii) \quad cars = \frac{17}{2} = 8.5$$

$$= 8$$

$$\left[\begin{array}{l} 1st \\ \text{car time} = 17 \end{array} \right]$$

$$\text{still cars remain} = (20 - 8)$$

$$= 12$$

Knowledge Test - 1 :

$$\text{Q4: a) users} = \frac{10 \times 10^6}{1 \times 10^6} \\ = 10$$

b) For no queuing delay, users ≤ 10
For queuing delay, users > 10 .

(c) a given users is transmitting.

$$P = (0.2)$$

$$\text{(d) } P = (0.2)(0.8)^{19} \\ = 2.88 \times 10^{-3}$$

$$\text{(e) } P = {}^{20}C_1 \cdot (0.2)^1 \cdot (0.8)^{19} \\ = 0.057$$

$$\text{(f) } P = (0.2)^{20} = 1.04 \times 10^{-14}$$

$$\text{(g) } P = \sum_{i=11}^{20} {}^{20}C_i (0.2)^i \cdot (0.8)^{20-i}$$

knowledge Test-2:

$$d_{\text{tran}} = \frac{L}{R}$$

$$d_{\text{prop}} = \frac{d}{s}$$

$$d_{\text{que}} = n \times \frac{L}{R}$$

$$\text{Q4: queuing delay} = (20 + 0.5) \times \frac{64 \times 8}{10 \times 10^6}$$
$$=$$

$$\text{Q5: It take} = \frac{1.45 \times 10^3}{10 \times 10^9} + \frac{500 \times 10^3}{2.5 \times 10^8}$$
$$= 2.001 \times 10^{-3} \text{ sec}$$

Ex-9 :

$$\text{For } R_1 = \frac{L}{R} + \frac{d}{S}$$

$$= \frac{4 \times 10^3 \times 8}{1 \times 10^6} + \frac{100 \times 10^3}{20000 \times 10^3}$$

$$\begin{aligned} \text{End-to-end} &= \frac{8 \times 10^3}{1 \times 10^6} \times 3 + \frac{100}{20000} \times 3 + 2 \times (5 \times 10^{-6}) \\ &= 0.039 \text{ sec} \end{aligned}$$

solve CT

$$\textcircled{1} \text{ Per slot} = \frac{60 \times 10^6}{30}$$

$$\text{time} = \frac{200000}{\frac{60 \times 10^6}{30}} = 0.1 + 21(N)$$

$$2(a) \frac{M}{R} = 10, \quad N \leq 10 : \text{Yes}$$

$N > 10 : \text{NO}$

$$(c) \sum_{i=11}^{50} {}^{50}C_i (0.4)^i (1-0.4)^{50-i} \approx \text{very small value}$$

(d) packet switching a better support than circuit switching. Jam probability on link is less, that's why packet switching better.

③

Layering Architecture

7 Application

6 Presentation

5 session

Application

Provides network specific services

4 Transport

Segmentation, process-to-process delivery

3 Network

Host-to-Host delivery

2 Data link

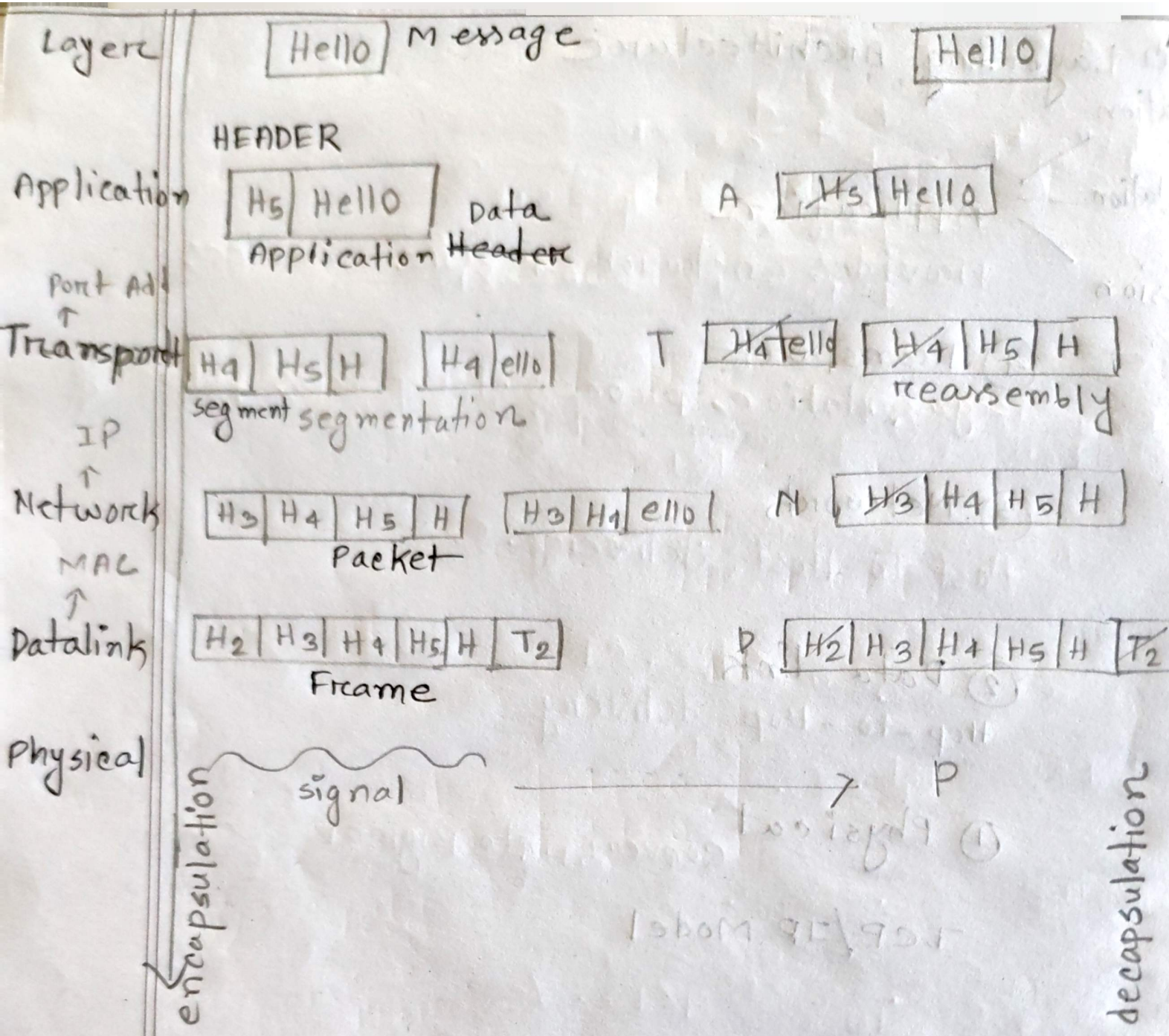
Hop-to-Hop delivery

1 Physical

Convert into signal

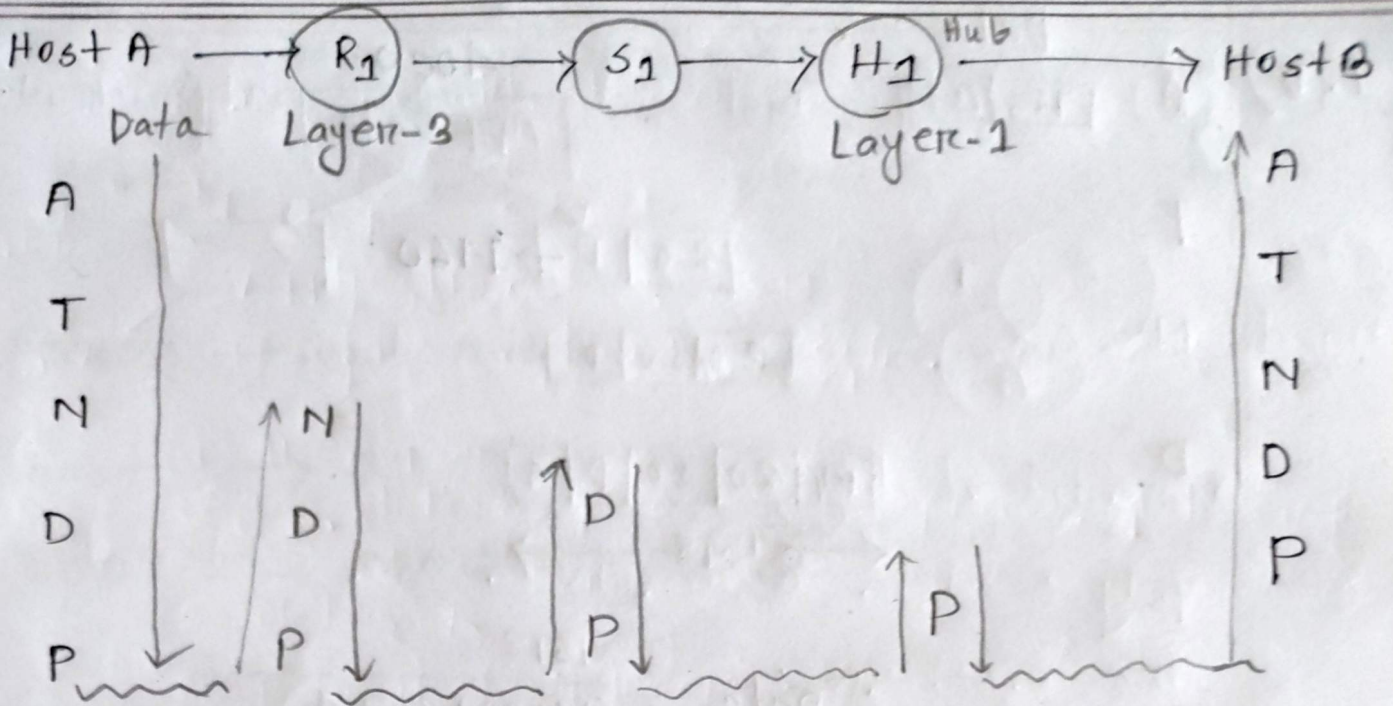


TCP/IP Model



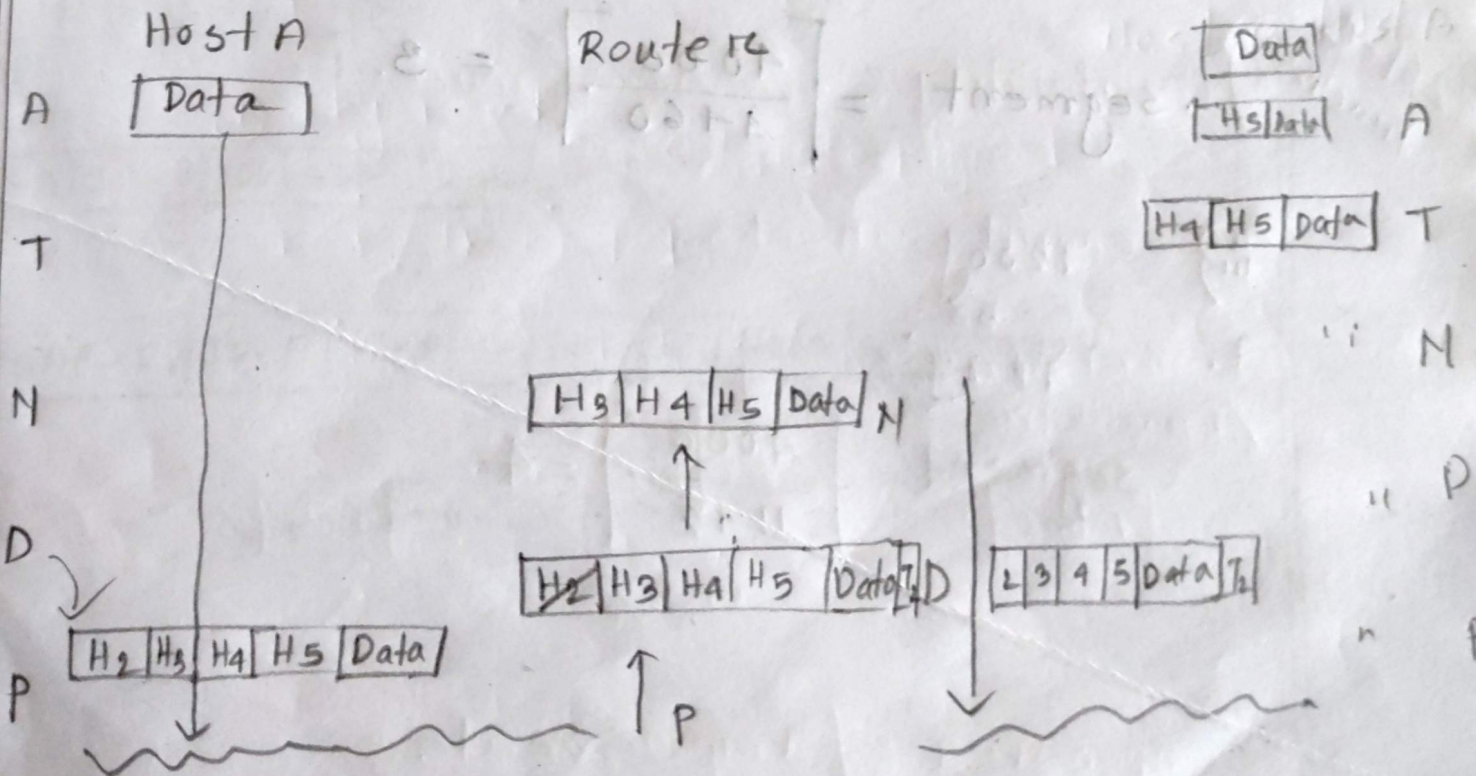
Segmentation depends on the datalink layer and is dependent on the physical layer.

Layer-2



261 - 1(b)

Host B



253 - (d)

1. (D)

A Data
[Hs | Data]

4000

App Data

T

[20 | D] → 1460

N

[20 | 20 | P]

D

[14 | 20 | 20 | D | 4]

P

← 1518 →

$$\therefore \text{Data} = \frac{1518}{(14+20+20+4)}$$

$$= 1460$$

1-C
practice

$$\therefore \text{segment} = \left\lceil \frac{4000}{1460} \right\rceil = 3$$

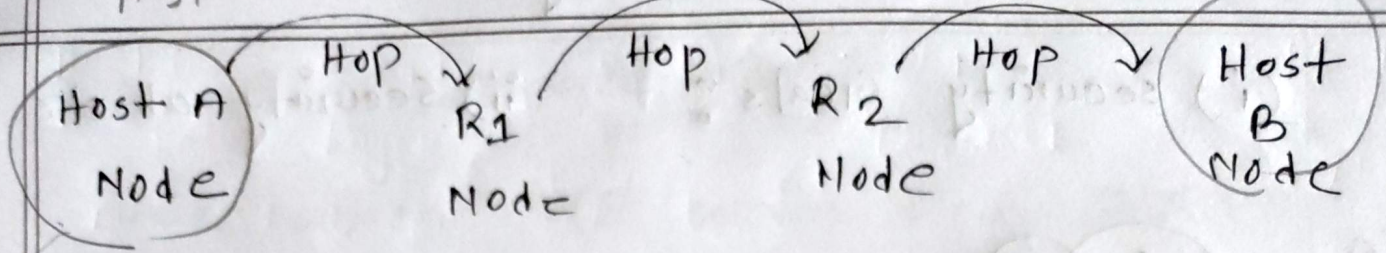
iii)

$$\begin{array}{r} 1200 \\ - 58 \\ \hline 1142 \end{array}$$

$$\therefore \text{segment} = \left\lceil \frac{4000}{1142} \right\rceil = 4$$

Host

Host



Process → Port Address

Host → IP Address

Hop → MAC Address [immediat next]

Data

HS | Data

100 | 50 | HS | Data

IP-A | IP-B | 100 | 50 | HS | Data

IP-A | IP-B | 100 | 50 | D

MAC-R1 | IP-A | IP-B | 100 | 50 | HS | Data

same

MACR1
MACR2